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Jaeyun (Jay) Ree

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LinkedIn,GitHub,Website

Final-year Software Engineering (Honours) student at Monash University, graduating Nov 2026, with research interests in quantum and classical algorithms for combinatorial optimisation. As a Research Assistant under Dr Mohammed Eunus Ali, I designed and benchmarked matching and assignment algorithms for autonomous-vehicle coordination, resulting in a short paper currently under review at ACM SIGSPATIAL 2026. I also completed a Human Research Ethics-approved Honours thesis on human-centred adaptive assessment. Alongside my research, I bring software engineering experience as co-founder and CTO of AIVA and through industry internships. I am a recipient of the 2023 and 2024 Monash Engineering Dean's Honours.

Publications & Research Output

Under review

- J. Ree, M. E. Ali, and M. A. Cheema, "Dynamic Platoon Formation: Spatio-Temporal Matching for Towing-Based Cooperative Autonomous Mobility" (short paper), submitted to the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026). *Under review.*

In preparation

- Manuscript in preparation extending my Honours thesis on human-centred adaptive assessment (comparing LLM-based and rule-based adaptive questionnaires), targeting submission in 2026.

Research Experience

Research Assistant – Monash University, Clayton Victoria Nov 2025 –
Project: Dynamic Platoon Formation of Multi-Type Autonomous Vehicles for Sustainable Urban Mobility

- Contributed to modelling the coordination task as a constrained optimisation problem with multi-segment matching and capacity constraints, under Dr. Mohammed Eunus Ali (with Prof. Aamir Cheema as co-author).
- Designed and implemented two matching algorithms — a Greedy Maximum-Weight Matching baseline and an Iterative Linear Assignment (ILA) method using the Jonker–Volgenant solver — achieving 25–52% energy-proportional distance coverage across synthetic scenarios with up to 400 passive and 80 active vehicles.
- Built an end-to-end experimental framework in **Python** (NumPy, SciPy, Pandas, Matplotlib) with parameter-sweep experiments, automated data pipelines, and comparative visualisation for algorithm benchmarking; revised and resubmitted the work across two conference formats in response to peer review.

Final Year Project Researcher (Honours Thesis) – Monash UniversityJul 2025 – Present
Human-Centred Adaptive Assessment (FIT4701 / FIT4702)

- Conducted a human-centred study comparing rule-based and LLM-based adaptive mental-health questionnaires, supervised by Dr ABM Russel with co-supervision from Prof. John Grundy (ARC Laureate Fellow) and industry mentorship from Mark Foley (Future Wellness Group).
- Designed and ran a MUHREC-approved within-subjects user study (Project ID 51664) with 40 participants, grounded in the COSMIN content-validity framework, and co-developed "Navigator", a Flutter/Supabase prototype implementing both adaptation methods.

- Performed quantitative analysis (paired-sample t-tests, effect sizes) and inductive thematic coding, and authored the final research report; the LLM-based method was rated significantly higher across all three evaluated dimensions (Clarity, Relevance, Quality).
- Continuing to extend the study beyond the completed coursework, with a manuscript in preparation targeting submission in 2026.

Professional Experience

Teaching Associate – Monash University, Melbourne Victoria **Mar 2026 – Jun 2026**

- Delivered weekly applied sessions for *FIT1045: Introduction to Programming*, mentoring students on C++ fundamentals, control structures, data types, and the SplashKit library; assessed portfolio submissions against structured rubrics.

Chief Technology Officer (CTO) – AIVA, Melbourne **Dec 2025 –**

- Co-founded and led end-to-end architecture of a full-stack AI executive-assistant platform: backend in Node.js (Express) with Supabase (PostgreSQL), mobile client in Flutter (iOS), and an autonomous AI agent (Google Gemini) for scheduling and task prioritisation.
- Directed technical strategy, sprint planning, and code review across frontend and backend teams, integrating Google Calendar, Notion, and Google Cloud voice APIs.

Full Stack Developer Intern – Job Search Works, Melbourne **May 2025 – Nov 2025**

- Built core features for an AI-powered job-application SaaS (resume–job matching, automated application workflows) using **React**, **TypeScript**, and **Firebase**; set up CI/CD with **GitHub Actions** and Husky.

IT Analyst – Hardware Co-Ordination Intern – Coles Group **Jul 2024 – Dec 2024**

- Coordinated hardware lifecycle management across Coles divisions; automated a manual reporting process with VBA (4 hours to 30 minutes) and contributed to over $90,000 in documented operational savings through improved vendor and lifecycle planning.

Education

Monash University **July 2020 – Present [Expected Graduation: Nov 2026]**

- Bachelor of Engineering (Honours) in Software Engineering; GPA 3.759 / 4.000, WAM 82.571 (High Distinction).
- Relevant coursework: Algorithms and Data Structures, Discrete Mathematics for Computer Science, Databases, Operating Systems, Computer Architecture. Proficient in **Python** (NumPy, SciPy, Pandas), **Java**, and **TypeScript/JavaScript**; experienced with Git, Linux, React, and SQL.

Monash College (Foundation Year) **July 2019 – June 2020**
GPA 4.0/4.0 (High Distinction)

Awards & Honours

2024 Monash Engineering Dean's Honours – Jun 2025
2023 Monash Engineering Dean's Honours – May 2024
Summer Vacation Research Scholarship, Faculty of IT – 2025
Industry-Based Learning Placement Scholarship – Jul 2024
Academic Commendation for Highest Grade in FIT2081: Mobile Application Development – May 2024
Engineering International Undergraduate Excellence Scholarship – Jun 2020

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Certifications

AWS Certified Cloud Practitioner – Mar 2025

Email: lee010518@gmail.com | **Mobile:** 0452134809 | **Referees:** Available on request